

No Finite Separable Representation at Three Prime Generators

*Irrational distinct-height sums for every finite set of primes, kernel rank and a tail
criterion for Erdős Problem #269*

WILL COOK*

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ABSTRACT

For every finite set P of at least two primes, the sum of the reciprocals of the distinct running least common multiples of the P -smooth integers is irrational. Erdős asserted this in a 1973 letter that gives no proof. For two primes it follows from a Hecke–Mahler transcendence theorem; the new case is three or more primes. A rational value would make the scaled tails of the sum integers, so two tails that are close would be equal. The tails after a prime power t and after tp^n , for a good return n , are close, and their words of primes differ by rearranging short runs; a point of a torus lying on exactly two coordinate walls produces a run of two primes whose rearrangement changes the tail. The proof is an ordinary one, refereed by a second reader. For $\{2, 3, 5\}$ a different proof, by five affine maps with disjoint images on dyadic blocks, is checked in Lean. The same finite-state method proves every set of three primes at most 31 by exact computation and fails for $\{2, 3, 5, 7\}$, where two different words of blocks have the same affine map. The same comparison of two tails proves that the sub-sum of the catalogue series over the powers of one prime is irrational. For three distinct primes the reciprocal running-LCM kernel has nonsingular minors of every order, so no finite sum of products separates one exponent from the other two. For the repeated $\{2, 3, 5\}$ series, which counts a running LCM once for every integer at which it occurs, we prove a tail recurrence and a residue criterion equivalent to irrationality. Its tails grow quadratically, the finite-state argument has no counterpart there, and the criterion’s inequalities for every multiplier coprime to 30 after arbitrarily late starts are unproved.

1 Introduction

For a finite prime set P , let $\mathcal{S}_P$ consist of the positive integers whose prime factors lie in P , including 1. We use $\mathbb{N} = \{0, 1, 2, \dots\}$ for exponent indices. For $x \geq 1$, put

$$L(x) = \text{lcm}\{u \in \mathcal{S}_P : u \leq x\}, \quad \mathcal{R}_P = \sum_{u \in \mathcal{S}_P} L(u)^{-1}.$$

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Erdős asked whether $\mathcal{R}_P$ is irrational when $|P| \geq 2$ [1, p. 65][2, p. 106]. For $P = \{p, q, r\}$, write

$$H(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor}, \quad K(i, j, k) = H(p^i q^j r^k)^{-1}.$$

Prime-power divisibility gives $L = H$ (Proposition 5.1). The running LCM is therefore constant between consecutive prime powers p^n, q^n, r^n ($n \geq 1$), and at each of them it is multiplied by the corresponding prime (Proposition 5.2). Counting each of its values once gives the *distinct-height sum*

$$\mathcal{D}_P = 1 + \sum_t H(t)^{-1},$$

where t runs over the prime powers p^n, q^n, r^n with $n \geq 1$. The two sums differ because the running LCM repeats: 5 and 6 both have running LCM 60, which contributes $2/60$ to $\mathcal{R}_{\{2,3,5\}}$ and $1/60$ to $\mathcal{D}_{\{2,3,5\}}$. In a letter dated 1 January 1973, Erdős called the irrationality of $\mathcal{R}_P$ a conjecture he could not prove and wrote that he could show the distinct-height sum to be irrational; the letter prints no argument [3]. Erdős and Graham state only the question for $\mathcal{R}_P$ [1, p. 65], and we have not found a proof in print (the companion's Section 11.1 lists the sources searched). Section 2 proves the irrationality of $\mathcal{D}_P$ for every finite P with $|P| \geq 2$.

Fan used the identity $L = H$ to separate the two-prime kernel [7]. With three primes, even a finite sum of separated products is impossible (Section 3). That theorem is about the kernel and says nothing about the arithmetic of its sum. Sections 5 and 6 turn to the repeated $\{2, 3, 5\}$ sum and state exactly what its irrationality still requires.

Formal proofs. A result with a kernel-checked Lean proof carries a mark in the margin. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. *Comparator* opens the record of an independent check, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean's Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text.

2 The distinct-height sums

Let P be a finite set of primes with $|P| \geq 2$ and put $H_P(t) = \prod_{r \in P} r^{\lfloor \log_r t \rfloor}$. The proof of Proposition 5.1 applies to any finite set of primes: the running LCM of the P -smooth integers up to t is $H_P(t)$, and it changes only at the prime powers r^n ($r \in P, n \geq 1$), where it is multiplied by r . List these prime powers as $t_1 < t_2 < \dots$, let q_k be the prime of t_k , and put $Q_k = q_1 \cdots q_k = H_P(t_k)$. Then

$$\mathcal{D}_P = 1 + \sum_{k \geq 1} \frac{1}{Q_k}.$$

For $P = \{2, 3, 5\}$ the prime powers $2, 3, 4, 5, 8, 9, 16, 25, 27, 32, \dots$ carry the heights $2, 6, 12, 60, 120, 360, 720, 3600, 10800, 21600, \dots$, so $\mathcal{D}_{\{2,3,5\}} = 1.7796169832947490504 \dots$

Theorem 2.1 (the distinct-height sums). *For every finite set P of primes with $|P| \geq 2$, the number $\mathcal{D}_P$ is irrational.*

Ordinary proof, given below. It was found within this project on 26 September 2026 and independently refereed by a second reader, whose five small corrections are incorporated. It has no Lean proof, so Comparator does not apply. The case $P = \{2, 3, 5\}$ is Theorem 2.6, which is checked in Lean.

For $|P| = 2$ the theorem is contained in Theorem 4.1: $\mathcal{D}_{\{p,q\}}$ is an affine function of one Hecke–Mahler value, which is transcendental by Loxton and van der Poorten [6, Theorem 8, p. 40] and Bugeaud and Laurent [5, Theorem 1.1]. The new case is $|P| \geq 3$, where every Hecke–Mahler theorem we found has a single slope.

The proof compares two tails of the series. A rational value makes every scaled tail an integer, so two tails closer than the scale are equal. The tails after t_k and after $t_k p^n$, for a suitable n , are that close. Their words of primes differ only by rearranging short runs, and equality of the tails forces an integer invariant f to agree on each run and its rearrangement. A run of two primes violates this, and a point of a torus on exactly two coordinate walls produces one.

Words. For an infinite word $u = u_1 u_2 \dots$ over P put $V(u) = \sum_{i \geq 1} (u_1 \dots u_i)^{-1}$. For a finite word $\sigma = \sigma_1 \dots \sigma_n$ put $\Pi(\sigma) = \sigma_1 \dots \sigma_n$ and

$$f(\sigma) = \sum_{t=1}^n \sigma_{t+1} \dots \sigma_n,$$

the term $t = n$ being 1. Then $f(\sigma)$ is a positive integer and

$$V(\sigma u) = \frac{f(\sigma) + V(u)}{\Pi(\sigma)}. \quad (1)$$

Put $x_k = V(q_{k+1} q_{k+2} \dots) = \sum_{j > k} Q_k / Q_j$. Every suffix of $q_1 q_2 \dots$ contains every prime of P , in particular one larger than $\min P$, so its value lies in $(0, 1/(\min P - 1)) \subseteq (0, 1)$.

Lemma 2.2 (integral tails). *If $\mathcal{D}_P = N/K$ with integers N and $K \geq 1$, then Kx_k is an integer for every $k \geq 1$.*

Proof. $Q_k \mathcal{D}_P = Q_k + \sum_{i \leq k} Q_k / Q_i + x_k$, and Q_i divides Q_k for $i \leq k$. So $Kx_k = Q_k N - K(Q_k + \sum_{i \leq k} Q_k / Q_i)$. $\square$

Lemma 2.3 (rearranged blocks). *Let $u = \sigma_1 \sigma_2 \dots$ and $u' = \sigma'_1 \sigma'_2 \dots$ be infinite words, where each σ_i is a nonempty finite word and σ'_i is a rearrangement of it, and suppose that every suffix of u or u' starting at a block boundary has value in $(0, 1)$.*

(i) *If $V(u) = V(u')$, then $f(\sigma_i) = f(\sigma'_i)$ for every i .*

(ii) *If $\sigma_i = \sigma'_i$ for every $i < i_0$, then $|V(u) - V(u')| < 1/\Pi(\sigma_1 \dots \sigma_{i_0-1})$.*

Proof. Let U_i and U'_i be the values of the suffixes starting at block i . Since $\Pi(\sigma_i) = \Pi(\sigma'_i)$, (1) gives $\Pi(\sigma_i)(U_i - U'_i) = f(\sigma_i) - f(\sigma'_i) + U_{i+1} - U'_{i+1}$. If $U_i = U'_i$, the integer $f(\sigma_i) - f(\sigma'_i)$ equals $U'_{i+1} - U_{i+1} \in (-1, 1)$, so both vanish. Induction from $U_1 = V(u) = V(u')$ proves (i). For (ii), $V(u) - V(u') = (U_{i_0} - U'_{i_0})/\Pi(\sigma_1 \dots \sigma_{i_0-1})$. $\square$

Lemma 2.4 (short rearrangements). *The map f is injective on the orderings of any set of at most three distinct primes.*

Proof. For two primes, $f(ab) = 1 + b$. For three, $f(\sigma_1\sigma_2\sigma_3) = 1 + \sigma_3(1 + \sigma_2)$. Let σ, σ' be orderings of $\{a, b, c\}$ with $f(\sigma) = f(\sigma')$. If $\sigma_3 = \sigma'_3$, then $\sigma_2 = \sigma'_2$ and $\sigma = \sigma'$. Otherwise say $\sigma_3 = a$ and $\sigma'_3 = b$, so $a(1 + \sigma_2) = b(1 + \sigma'_2)$ with $\sigma_2 \in \{b, c\}$ and $\sigma'_2 \in \{a, c\}$. The pairs (b, a) and (c, c) give $a = b$, the pair (b, c) gives $b \mid a$, and (c, a) gives $a \mid b$. $\square$

For four primes injectivity fails: $f(5, 7, 3, 2) = f(7, 2, 3, 5) = 51$. The proof therefore looks for a run of two primes, for which $f(ab) - f(ba) = b - a \neq 0$.

The torus. Let $\mathbb{T} = \prod_{q \in P} \mathbb{R}/(\log q)\mathbb{Z}$ and $\Theta(s) = (s \bmod \log q)_{q \in P}$ for $s \in \mathbb{R}$. For $\theta \in \mathbb{T}$ a q -crossing is a real t with $\theta_q + t \in (\log q)\mathbb{Z}$, and $W(\theta)$ is the word of the primes of the positive crossings in increasing order of time. For $\theta_k = \Theta(\log t_k)$ the positive q -crossings are the times $\log(q^j/t_k)$ with $q^j > t_k$, so $W(\theta_k) = q_{k+1}q_{k+2}\dots$, and crossings of different primes never coincide.

Let T_0 be the closure of $\Theta(\mathbb{R})$, a subtorus containing every θ_k . For $t > 0$ and $\varepsilon > 0$, the pigeonhole principle applied to the points $\Theta(jt')$, $j \geq 1$, with $t' > t$, gives $d > t$ with $\Theta(d)$ within ε of 0, so $\Theta(-t)$ is within ε of $\Theta(d - t)$. Hence $\Theta([0, \infty))$ is dense in T_0 , and so is every forward orbit $\{\theta + \Theta(t) : t \geq t_0\}$ with $\theta \in T_0$.

In the coordinates $\varphi_q = \theta_q / \log q$ the torus is $\mathbb{R}^P / \mathbb{Z}^P$, and by Kronecker's theorem T_0 is the image of the smallest rational subspace $L \subseteq \mathbb{R}^P$ containing $\omega = (1/\log q)_{q \in P}$. The annihilator of L is a rational subspace spanned by the integer vectors c with $\sum_q c_q / \log q = 0$. If it contained a nonzero vector supported on at most two primes a, b , it would contain an integer one, and $\log a / \log b$ would be rational, which is false because no power of a is a power of b . So each φ_q is nonzero on L and any two are linearly independent on L . In particular the linear form $\theta_b - \theta_a = (\log b)\varphi_b - (\log a)\varphi_a$ is nonzero on L , although it vanishes on the flow direction ω .

Lemma 2.5 (two walls). *If $|P| \geq 4$, there is $x \in T_0$ with $\theta_q(x) = 0$ for exactly two primes $q \in P$.*

Proof. For $q \neq q'$ the function $(u_1, u_2) \mapsto \varphi_q(u_1)\varphi_{q'}(u_2) - \varphi_q(u_2)\varphi_{q'}(u_1)$ is a nonzero polynomial on $L \times L$, since φ_q and $\varphi_{q'}$ are independent on L . Rational points are dense in $L \times L$, so there are rational $u_1, u_2 \in L$ at which all these determinants are nonzero, and a change of basis of their span multiplies every determinant by the same nonzero constant. Their span is a rational plane; with a $\mathbb{Z}$ -basis e_1, e_2 of its integer points, $s \mapsto s_1e_1 + s_2e_2$ identifies $\mathbb{R}^2/\mathbb{Z}^2$ with a subtorus $S \subseteq T_0$. On S the condition $\theta_q = 0$ reads $\langle \gamma_{q'}, s \rangle \in \mathbb{Z}$ with $\gamma_q = (\varphi_q(e_1), \varphi_q(e_2)) \in \mathbb{Z}^2$, and these vectors are nonzero and pairwise non-parallel. Each set $\{\theta_q = 0\} \cap S$ is therefore a family of finitely many disjoint parallel closed geodesics.

Two non-parallel closed geodesics of the flat torus meet. So the union of the families is a graph whose vertices are the points on at least two families, and every geodesic contains a vertex. Let k_v be the number of families through a vertex v . A geodesic with r vertices is cut into r edges, and geodesics of one family are disjoint, so $E = \sum_v k_v$. For $q \neq q'$ the map $s \mapsto (\langle \gamma_q, s \rangle, \langle \gamma_{q'}, s \rangle)$ is a covering of $\mathbb{R}^2/\mathbb{Z}^2$ of degree $|\det(\gamma_q, \gamma_{q'})|$, so each component of the complement of these two families is an open parallelogram.

Each face of the arrangement lies in such a parallelogram, where the other families are straight chords, so it is an open convex polygon with at least three sides. Euler's formula $V - E + F = 0$ for the torus and $2E \geq 3F$ give $\sum_v (k_v - 3) \leq 0$. The origin lies on all $|P| \geq 4$ families, so some vertex has $k_v = 2$, and it is the required x . $\square$

Proof of Theorem 2.1. Suppose $\mathcal{D}_P = N/K$. Fix $p \in P$ and an index k with $q_k = p$. If $|P| \geq 4$, fix a, b and x as in Lemma 2.5, and put $\eta = \min(1/10, \min_{c \neq a, b} \text{dist}(\theta_c(x), 0)) > 0$, distances being taken in $\mathbb{R}/(\log c)\mathbb{Z}$. Let $\beta = \min_{q \neq p} \text{dist}(\theta_{k,q}, 0)$, which is positive because t_k is a power of p . Choose $T > 0$ such that $2^\nu \geq K$, where ν is the number of crossings of θ_k in $(0, T]$, and let $g > 0$ be the least distance between two crossings of θ_k in $[0, T + 1]$.

For $n \geq 1$ let $t_m = t_k p^n$, so that $\theta_m = \theta_k + \Theta(n \log p)$. Write $\delta_p = 0$ and $\delta_q = n \log p - m_q \log q$ for $q \neq p$, with m_q the nearest integer to $n \log p / \log q$; then $\Theta(n \log p)$ has coordinates $\delta_q \bmod \log q$. By Dirichlet's simultaneous approximation theorem we may choose n with

$$\varepsilon = \max_q |\delta_q| < \min\left(\beta, \frac{g}{2}, \frac{\log 2}{2|P|}, \frac{\eta}{8}\right),$$

the last term only when $|P| \geq 4$. Then every $m_q \geq 1$, and the numbers δ_q are pairwise distinct because p^n and the q^{m_q} are pairwise distinct.

Chains. The q -crossings of θ_m are those of θ_k shifted by $-\delta_q$. Every q -crossing s of θ_k with $q \neq p$ has $|s| \geq \beta > |\delta_q|$, so the shift preserves signs and matches the positive q -crossings of θ_k and θ_m in order. Cut the positive crossings of θ_k wherever two consecutive ones are more than 2ε apart, and call the pieces *chains*. Crossings in different chains are more than 2ε apart and their shifts differ by less than 2ε , so they keep their order. Hence $W(\theta_m)$ is the concatenation of the chains of $W(\theta_k)$ in the same order, each rearranged. Two crossings of one prime are at least $\log 2$ apart, and any $|P| + 1$ consecutive crossings of one chain lie within $2\varepsilon|P| < \log 2$; they would belong to $|P| + 1$ distinct primes, which is impossible. So a chain has at most $|P|$ crossings, and they belong to distinct primes.

Equal tails. A chain meeting $(0, T]$ is a single crossing, since the crossings in $[0, T + 1]$ are at least $g > 2\varepsilon$ apart and a crossing beyond $T + 1$ is more than 1 away. So the first ν blocks of the two chain decompositions agree, and Lemma 2.3(ii) gives $|x_k - x_m| < 2^{-\nu} \leq 1/K$. By Lemma 2.2, Kx_k and Kx_m are integers less than 1 apart, so $x_k = x_m$. Lemma 2.3(i) then gives $f(C) = f(C')$ for every chain C of $W(\theta_k)$ and its rearrangement C' in $W(\theta_m)$.

Case $|P| \leq 3$. Take $q \neq p$, and suppose $\delta_q > 0$; the other sign is symmetric. The positive p -crossings of θ_k are the numbers $j \log p$ with $j \geq 1$, and its positive q -crossings are the numbers $i \log q - \log t_k$ with i large. Since $\log p / \log q$ is irrational, the differences $j \log p - i \log q + \log t_k$ with i, j large are dense in $\mathbb{R}$. So some p -crossing $j \log p$ and q -crossing s of θ_k satisfy $s - \delta_q < j \log p < s$. Their shifts reverse their order, so they lie in one chain, which is genuinely rearranged in $W(\theta_m)$. It consists of at most three distinct primes, and Lemma 2.4 gives $f(C) \neq f(C')$, a contradiction.

Case $|P| \geq 4$. Put $D = \delta_a - \delta_b$, so $0 < |D| < 2\varepsilon$. Since $\theta_b - \theta_a$ is a nonzero linear form on L , there is $y \in L$ with $|\theta_q(y)| < \eta/8$ for every q and $\theta_b(y) - \theta_a(y)$ strictly between 0 and D . Let z be the point $x + y + \Theta(-\eta/2)$ of T_0 , with y read in the torus. The a -crossings

of z are the times $\eta/2 - \theta_a(y) + j \log a$, so exactly one of them, τ_a , lies in $[-\eta/4, 5\eta/4]$, and it lies in $(\eta/4, 3\eta/4)$; the same holds for b , and $\tau_a - \tau_b$ lies strictly between 0 and D . Every c -crossing of z with $c \neq a, b$ is at distance at least $7\eta/8$ from $\eta/2$, so it lies outside $[-\eta/4, 5\eta/4]$. Let U be the set of $\theta \in T_0$ with exactly one a -crossing and one b -crossing in $[-\eta/4, 5\eta/4]$, both in $(\eta/4, 3\eta/4)$, with their difference $\tau_a - \tau_b$ strictly between 0 and D , and with no other crossing in $[-\eta/4, 5\eta/4]$. Each condition is open, so U is an open subset of T_0 containing z , and by density $\theta_k + \Theta(t^*) \in U$ for some $t^* > T + 2$.

Then θ_k has an a -crossing $t^* + \tau_a$ and a b -crossing $t^* + \tau_b$ less than $|D| < 2\varepsilon$ apart, and every other crossing of θ_k is more than $\eta/2 > 4\varepsilon$ away from both. So these two crossings form a chain C . Their shifts differ by $(\tau_a - \tau_b) - D$, whose sign is opposite to that of $\tau_a - \tau_b$, so C' is C reversed. But $f(ab) - f(ba) = b - a \neq 0$, a contradiction. $\square$

Written as the Cantor series $\sum_{k \geq 0} (q_1 \cdots q_k)^{-1}$, $\mathcal{D}_P$ has every digit equal to 1 and every base in P . The irrationality criteria of Erdős–Straus [9, Theorem 2.1] and Hančl–Tijdeman [10, Theorem 3.1] assume that each digit divided by the product of the last two bases in its denominator tends to 0; here that quotient is at least $(\max P)^{-2}$, so neither applies. The rationality criterion of Diananda and Oppenheim [8] condenses a Cantor series into blocks of consecutive terms, the form of the dyadic argument below; it restates rationality and does not by itself decide $\mathcal{D}_P$.

2.1 The case $\{2, 3, 5\}$, checked in Lean

Theorem 2.6 (the distinct-height sum for $\{2, 3, 5\}$). *The number $\mathcal{D}_{\{2,3,5\}}$ is irrational.*

Lean

Checked in Lean for the sum exactly as defined in the introduction ([Lean proof](#)); Comparator has not yet been run on it. The constants of the proof below, the five pairs (b_τ, μ_τ) , the bounds $3/10$ and $13/15$ and the four gaps, are also recomputed in exact arithmetic by the [distinct-height programs](#).

This is a case of Theorem 2.1. The Lean proof follows a different route, which reads the sum one dyadic block at a time. Between consecutive powers of two there is at most one power of 3 and at most one power of 5, so each block acts on the normalised tail by one of five affine maps, and the images of the possible tails under these maps are disjoint. A tail therefore determines the block before it, and with it every later block. A rational value would make two tails coincide and the pattern of blocks repeat, and the powers of 3 do not fall periodically among the dyadic blocks.

Proof. List the prime powers $2^n, 3^n, 5^n$ ($n \geq 1$) in increasing order as $t_1 < t_2 < \dots$. For $a \geq 0$ the block $(2^a, 2^{a+1}]$ ends with the jump at 2^{a+1} . Consecutive powers of 3, and of 5, have ratio greater than 2, so the block contains at most one power of each odd prime, both strictly inside it. Its type τ_a is the word formed by those odd primes in increasing order of their powers, one of $\emptyset, 3, 5, 35, 53$; for instance $(16, 32]$ contains $25 < 27$, so $\tau_4 = 53$.

Put $P_a = H(2^a)$ and $Y_a = P_a \sum_{t_k > 2^a} H(t_k)^{-1}$. If block a has interior primes $r_1, \dots, r_v$ in this order, its jumps have heights $P_a r_1, P_a r_1 r_2, \dots, P_a r_1 \cdots r_v$ and finally $P_{a+1} = 2P_a r_1 \cdots r_v$. Splitting off the block gives

$$Y_a = G_{\tau_a}(Y_{a+1}), \quad G_\tau(y) = \frac{\mu_\tau + y}{b_\tau}, \quad b_\tau = 2r_1 \cdots r_v, \quad \mu_\tau = 1 + b_\tau \sum_{j \leq v} \frac{1}{r_1 \cdots r_j}. \quad (2)$$

For $\tau = \emptyset, 3, 5, 35, 53$ the pairs (b_τ, μ_τ) are $(2, 1)$, $(6, 3)$, $(10, 3)$, $(30, 13)$ and $(30, 9)$.

Every jump multiplies the height by at least 2, so $0 < Y_a \leq 1$, and each G_τ is increasing and maps $[0, 1]$ into itself. By (2), $Y_a \geq \mu_{\tau_a}/b_{\tau_a} \geq 3/10$. The power 5^f lies in block $\lfloor f \log_2 5 \rfloor$ and $2 < \log_2 5 < 3$, so the blocks containing a power of 5 begin with block 2 and are two or three apart; every three consecutive blocks include one. On $[0, 1]$ a map whose block contains a power of 5 is at most $G_{35}(1) = 7/15$. So if $a + i$, with $i \leq 2$, is the first block from a onwards containing a power of 5, then $Y_{a+i} \leq 7/15$, and the at most two maps leading back to Y_a are $G_\emptyset$ or G_3 , both bounded by $G_\emptyset$ because $G_\emptyset(y) - G_3(y) = y/3 \geq 0$. Hence

$$\frac{3}{10} \leq Y_a \leq G_\emptyset(G_\emptyset(7/15)) = \frac{13}{15} \quad (a \geq 0).$$

The five maps send $I = [3/10, 13/15]$ to the intervals

$$\left[\frac{31}{100}, \frac{74}{225}\right], \left[\frac{33}{100}, \frac{29}{75}\right], \left[\frac{133}{300}, \frac{104}{225}\right], \left[\frac{11}{20}, \frac{29}{45}\right], \left[\frac{13}{20}, \frac{14}{15}\right] \quad (\tau = 53, 5, 35, 3, \emptyset),$$

separated by the gaps $1/900$, $17/300$, $79/900$ and $1/180$. Since $Y_{a+1} \in I$, the value Y_a lies in the image of τ_a and of no other type. Thus Y_a determines τ_a , then $Y_{a+1} = b_{\tau_a} Y_a - \mu_{\tau_a}$, and by induction the whole word $\tau_a \tau_{a+1} \dots$.

Suppose $\mathcal{D}_{\{2,3,5\}} = N/K$ with integers N and $K \geq 1$. For $t \leq 2^a$ the height $H(t)$ divides P_a , so

$$P_a \mathcal{D}_{\{2,3,5\}} - Y_a = P_a \left(1 + \sum_{t_k \leq 2^a} H(t_k)^{-1}\right) \in \mathbb{Z},$$

and KY_a is an integer in $[3K/10, 13K/15]$ for every $a \geq 0$. Two indices $a < a'$ give the same integer. Then $Y_a = Y_{a'}$, so $\tau_{a+j} = \tau_{a'+j}$ for every $j \geq 0$, and the word (τ_a) is eventually periodic with period $\ell = a' - a$.

Finally, $2^a < 3^e < 2^{a+1}$ holds exactly when $a\alpha < e < (a+1)\alpha$, where $\alpha = \log 2 / \log 3$ is irrational. Block a therefore contains $\lfloor (a+1)\alpha \rfloor - \lfloor a\alpha \rfloor$ powers of 3, a number recorded by τ_a . Summing over one period, the difference $\lfloor (a+\ell)\alpha \rfloor - \lfloor a\alpha \rfloor$ is a constant c for all large a , hence $\lfloor (a+n\ell)\alpha \rfloor = \lfloor a\alpha \rfloor + nc$ for every $n \geq 1$. Dividing by n and letting $n \rightarrow \infty$ gives $\ell\alpha = c$, that is $2^\ell = 3^c$, which is impossible. $\square$

The upper bound is the delicate step. Since $G_3(y) = G_\emptyset(y/3)$ and $G_{53}(y) = G_5(y/3)$, the images of the empty block and of the block with one power of 3, and those of the blocks 5 and 53, are disjoint exactly when the largest tail is less than three times the smallest, and the trivial bound $Y_a \leq 1$ would not separate them; the spacing of the powers of 5 lowers it to $13/15$.

2.2 A finite-state method, and where it stops

The dyadic argument extends to other prime sets through a finite automaton. It gives an independent proof for every set of three primes at most 31 and for most sets of four, and it fails at $\{2, 3, 5, 7\}$ for a reason the proof of Theorem 2.1 explains.

Corollary 2.7 (three primes up to 31). *For every set P of three primes, each at most 31, the distinct-height sum $\mathcal{D}_P$ is irrational.*

A case of Theorem 2.1, proved here by the independent finite-state method: an ordinary proof of the criterion below, applied through an exact finite computation for each of the 165 sets; no Lean proof. The long record proves the criterion in full.

Proof. Let $P = \{p < q < r\}$ and group the prime powers into the blocks $(p^a, p^{a+1}]$. Each block ends with p^{a+1} and contains at most one power of q and one of r , so, as in (2), the tails follow one of five increasing affine maps, one for each ordered set of interior primes. The power q^f lies in block $[f \log_p q]$, so consecutive blocks containing a power of q are g_q or $g_q + 1$ apart, where $p^{g_q} < q < p^{g_q+1}$, and likewise for r . Record before each block the number of blocks since q and since r last appeared. These pairs are the states of a finite automaton, and once both primes have appeared the actual sequence of blocks is one of its infinite paths. Start with the interval $[0, 1/(p-1)]$, which contains every tail, at every state, and repeatedly replace the interval at a state by the smallest interval containing the images of the intervals at its successors; every tail at that state stays inside. Suppose that for some k the images of these intervals under all compositions of k maps along the automaton, grouped by the first map, are pairwise disjoint. Then a tail determines its block, and the rest of the proof of Theorem 2.6 applies with $[0, 1/(p-1)]$ in place of I and $\log p / \log q$ in place of α . An exact computation, with interval endpoints rounded outwards to multiples of 2^{-96} , finds such a $k \leq 4$ for each of the 165 sets: $k = 1$ for 141 of them, and $k = 4$ for $\{2, 17, 31\}$, $\{2, 23, 29\}$ and $\{2, 29, 31\}$. A second computation, which solves exactly for the least and greatest values of the infinite paths from each state, returns the same k for every set. $\square$

For the 330 sets of four primes at most 31 the test succeeds for 292 at some depth $k \leq 5$, for instance at $k = 1$ for $\{3, 5, 7, 11\}$. For $\{2, 3, 5, 7\}$ it fails at every depth. The two-block words $(5, 7, 3), \emptyset$ and $(7), (3, 5)$ have the same map,

$$G_{573} \circ G_{\emptyset}(y) = G_7 \circ G_{35}(y) = \frac{103 + y}{420},$$

which is the collision $f(5, 7, 3, 2) = f(7, 2, 3, 5) = 51$ read in blocks. Both words continue by the blocks $(3), (5, 7)$ to the same state of the automaton, so the images for the first types $(5, 7, 3)$ and (7) overlap at every depth, and a single tail cannot determine its block. The proof of Theorem 2.1 compares two tails, and there one run of two primes suffices.

3 Nonsingular minors of every order

Theorem 3.1 (no finite separation of the kernel). *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 0$ there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(K(I(a), J(b), k))_{0 \leq a, b < n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell(i)$ and $G_\ell(j, k)$, $0 \leq \ell < d$, satisfying

$$K(i, j, k) = \sum_{\ell < d} f_\ell(i) G_\ell(j, k) \quad \text{for all } i, j, k.$$

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Proof. For fixed k , we rescale rows and columns to obtain a matrix independent of k . Put $c = r^{-1}$, $x_i = \{i \log_r p\}$ and $y_j = \{j \log_r q\}$, where braces denote fractional parts. Splitting the floor exponents gives

$$K(i, j, k) = U_i(k)^{-1} C_{ij} V_j(k)^{-1}, \quad C_{ij} = c^{\mathbf{1}_{\{x_i + y_j \geq 1\}}},$$

where

$$U_i(k) = p^i q^{\lfloor \log_q(p^i r^k) \rfloor} r^{k + \lfloor \log_r p^i \rfloor},$$

$$V_j(k) = q^j p^{\lfloor \log_p(q^j r^k) \rfloor} r^{\lfloor \log_r q^j \rfloor}.$$

The positive factors do not affect whether a minor vanishes; C records only whether $x_i + y_j \geq 1$. Distinct primes make $\log_r p$ and $\log_r q$ irrational, so each fractional-part orbit is dense in $[0, 1]$. The row and column indices are independent choices; no density of a single orbit of pairs is required. For $n \geq 1$, choose distinct rows with $0 < x_{I(0)} < \dots < x_{I(n-1)} < 1$. We choose each column so that its entries change from 1 to c at a different selected row. Writing $s_b = 1 - y_{J(b)}$, density of (y_j) lets us choose

$$s_0 \in (0, x_{I(0)}), \quad s_b \in (x_{I(b-1)}, x_{I(b)}) \quad (1 \leq b < n).$$

The intervals for the s_b are disjoint, so the column indices are distinct. The strict inequalities also avoid the threshold itself, and give $x_{I(a)} + y_{J(b)} \geq 1$ exactly when $b \leq a$. Thus multiplying row a by $U_{I(a)}(k)$ and column b by $V_{J(b)}(k)$ reduces the selected kernel minor to

$$T_n(c) = \begin{pmatrix} c & 1 & \dots & 1 \\ c & c & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ c & c & \dots & c \end{pmatrix}, \quad \det T_n(c) = c(c-1)^{n-1}.$$

Indeed, subtracting each preceding row from the next, working upwards from the last row, leaves diagonal entries $c, c-1, \dots, c-1$. The original determinant is therefore

$$c(c-1)^{n-1} \prod_{a < n} U_{I(a)}(k)^{-1} \prod_{b < n} V_{J(b)}(k)^{-1} \neq 0.$$

The indices were chosen from C , independently of k . For $n = 0$ the empty determinant equals 1.

For the final assertion, suppose a separation with d summands existed and fix any k . On the rows $I(0), \dots, I(d)$ and columns $J(0), \dots, J(d)$ its matrix would be the product of the $(d+1) \times d$ matrix $(f_\ell(I(a)))_{a,\ell}$ and the $d \times (d+1)$ matrix $(G_\ell(J(b), k))_{\ell,b}$. Its rank would be at most d , so its determinant would vanish, contrary to the minor just constructed. $\square$

No continuity or boundedness is assumed for the separated factors.

Corollary 3.2 (the same minors modulo integers coprime to 30). *For $(p, q, r) = (2, 3, 5)$ and every $n \geq 1$, there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$, chosen independently of B and k , such that for every $B \geq 2$ coprime to 30 and every $k \geq 0$, the selected $n \times n$ kernel matrix has unit determinant over $\mathbb{Z}/B\mathbb{Z}$, with each reciprocal prime power interpreted by its modular inverse.*

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Proof. Choose the maps from Theorem 3.1. Every row and column factor is a unit modulo B . The normalised determinant is $5^{-1}(-4/5)^{n-1}$, also a unit. $\square$

The modulus may be composite, for example 49 or 77. Coprimality with 30 makes both the reciprocal entries and the determinant units: the latter introduces only a power of 4 in its numerator.

Example 3.3. For $(p, q, r) = (2, 3, 5)$, the leading two-by-two determinant is $-1/15$: its four entries are $1, 1/6, 1/2, 1/60$, so it equals $1/60 - 1/12$.

The indices must be selected. The leading 4×4 block at $\{2, 3, 5\}$ is singular:

$$K(3, j, 0) = \frac{1}{120} K(0, j, 0) \quad (0 \leq j < 4).$$

These equalities follow by evaluating the height at 3^j and $8 \cdot 3^j$. But at $j = 4$, $K(3, 4, 0) - K(0, 4, 0)/120 = -1/19440000$. Thus the leading minors do not supply the arbitrary-order theorem.

Proposition 3.4 (rank of a matrix of threshold columns). *Let $m \geq 1$ and let c lie in a field with $c \neq 0, 1$. For $0 \leq k \leq m$ write v_k for the length- m column with a 1 in each of the first k coordinates and c thereafter. A matrix whose distinct columns are v_k for k in a nonempty set E has rank $|E| - \mathbf{1}_{\{0, m\} \subseteq E}$.*

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The correction accounts for the two constant columns: $v_0 = cv_m$. There is no other dependence among distinct threshold columns.

Proof. List $E = \{k_1 < \dots < k_t\}$. The $t - 1$ differences $v_{k_{j+1}} - v_{k_j} = (1 - c)\mathbf{1}_{\{k_j, \dots, k_{j+1}-1\}}$ have disjoint nonempty supports, hence are linearly independent. If $k_1 > 0$ or $k_t < m$, their union misses a coordinate where v_{k_1} is nonzero, so the rank is t . If $k_1 = 0$ and $k_t = m$, then $v_0 = c\mathbf{1}$ lies in the span of the differences (their sum is $(1 - c)\mathbf{1}$), so the rank is $t - 1$. $\square$

The restrictions $c \neq 0, 1$ exclude a zero column or identical columns; they hold for $c = 1/r$ over $\mathbb{Q}$. In one fixed layer k , ordering the sampled row phases puts every column in the stated form. The formula therefore determines the rank of each nonempty rectangular sample. For fixed row and column indices this rank is independent of k . Section 5 of the companion paper, *The Three-Prime Running LCM: Kernel Rank and Tail Arithmetic*, gives the exact rational comparisons.

Exact rank is not an approximation obstruction. Truncating the original kernel to $i < N$ gives a sum of N separated terms. The bound $H(x) > x^3/(pqr)$ makes the truncation error tend to zero both uniformly and in $\ell^1(\mathbb{N}^3)$. By contrast, the rescaled matrix in the rank proof has uniform distance $(1 - 1/r)/2$ from the matrices of finite rank. Rescaling preserves exact rank but removes the decay responsible for the first approximation. The companion proves both assertions, including the distinction between finite restrictions and the full infinite matrix, in Section 5.1.

4 The two-prime comparison

Fan's post of 26 June 2026 gives the repeated two-prime factorisation, its quadratic expression in one Hecke–Mahler value and its transcendence conclusion [7]. Counting the contributions at powers of each prime also gives the affine formula for the distinct-height sum below. The calculation fixes the normalisations and distinguishes the two sums.

Theorem 4.1 (both two-prime sums). *Let $p < q$ be distinct primes. Put $\theta = \log p / \log q$ and $A = \sum_{n \geq 0} p^{-n} q^{-\lfloor n\theta \rfloor}$. Let $\mathcal{R}_{p,q}$ sum the reciprocal running LCM at every positive $\{p, q\}$ -smooth integer, and let $\mathcal{D}_{p,q}$ count each distinct running LCM once. Then*

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$$\mathcal{D}_{p,q} = \frac{(q-p)A + p}{q-1}, \quad \mathcal{R}_{p,q} = \frac{(p+q-1)A - (p-1)A^2}{q-1}. \quad (3)$$

Both numbers are transcendental.

The Lean proof assumes the transcendence theorem of Bugeaud and Laurent; the two identities in (3) are proved in Lean without it.

Proof. Set $x = 1/p$, $y = 1/q$, $m_n = \lfloor n\theta \rfloor$ and $\delta_n = m_{n+1} - m_n$. Unique factorisation makes θ irrational, and $0 < \theta < 1$ gives $\delta_n \in \{0, 1\}$. The initial value and the positive powers of p contribute $A = \sum_{n \geq 0} x^n y^{m_n}$. There is one q -power strictly between p^n and p^{n+1} exactly when $\delta_n = 1$, so the positive powers of q contribute

$$B = \sum_{n \geq 0} \delta_n x^n y^{m_{n+1}}, \quad \mathcal{D}_{p,q} = A + B.$$

Since $0 < x, y < 1$, these series converge absolutely. The identity $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n} (y - 1)$ and an index shift in A give $A - 1 - xA = x(y - 1)B/y$. Therefore

$$B = \frac{p - (p-1)A}{q-1}.$$

At $p^i q^j$ the height is $p^{i+\lfloor j/\theta \rfloor} q^{j+m_i}$, hence

$$\mathcal{R}_{p,q} = A \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor} = A(1 + B).$$

For the last equality, each $j \geq 1$ corresponds to $n = \lfloor j/\theta \rfloor$ with $m_n = j - 1$ and $\delta_n = 1$. These identities prove (3).

For the Hecke–Mahler series $F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{m_n} x^n y^k$, geometric summation gives

$$A = \frac{1}{1-x} - \frac{1-y}{y} F_\theta(x, y). \quad (4)$$

Bugeaud and Laurent's theorem [5, Theorem 1.1] applies with intercept $\rho = 0$, $\beta = x$ and $\alpha = y$: the slope θ is irrational and lies in $(0, 1)$, x and y are nonzero algebraic numbers, $|x| < 1$ and $|xy^\theta| = p^{-2} < 1$. Hence $F_\theta(x, y)$ is transcendental; this case $\rho = 0$ is due to Loxton and van der Poorten [6, Theorem 8, p. 40]. Thus A is transcendental. The displayed affine and quadratic polynomials are nonconstant, so an algebraic value of either would force A to be algebraic. $\square$

The same calculation applies to the monoid generated by any coprime integers $1 < p < q$, such as 4, 9, but need not describe all integers supported on their prime factors. The companion's Section 3 gives this extension and the failure for 4, 8.

The distinct-height sums for every finite set of primes were treated in Section 2. The tail argument of the next two sections starts again from the running LCM and retains every multiplicity; it does not use the rank theorem.

5 The recurrence for the repeated sum

Proposition 5.1 (the running least common multiple). *Let p, q, r be pairwise distinct primes and $x \geq 1$. Then the running LCM equals the three-prime height: $L(x) = H(x)$.*

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Proof. Every smooth $n \leq x$ has prime exponents bounded by the corresponding integer logarithms, so $n \mid H(x)$. Conversely the three maximal pure powers occur among those smooth numbers; their product divides the running LCM because they are pairwise coprime. $\square$

The inequalities $x/p < p^{\lfloor \log_p x \rfloor} \leq x$ for each of the three primes also give

$$x^3/(pqr) < H(x) \leq x^3. \quad (5)$$

Proposition 5.2 (cells and jumps). *The running LCM is constant when the three integer logarithms are constant. A jump in exactly one logarithm multiplies it by the corresponding prime. The first n positive powers of each prime, together with 1, form $3n + 1$ distinct points.*

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Proof. The first two claims follow from the height formula. Positive powers of different primes cannot coincide, by unique factorisation, and none is 1. $\square$

For $(p, q, r) = (2, 3, 5)$ the smooth numbers 5 and 6 share the height 60, as noted in the introduction. The next identity retains such multiplicities when equal heights are grouped.

Proposition 5.3 (grouping terms with the same height). *For a finite exponent box $\mathcal{B}$, set $F(H) = \{(i, j, k) \in \mathcal{B} : H(p^i q^j r^k) = H\}$. Then*

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$$\sum_{(i,j,k) \in \mathcal{B}} K(i, j, k) = \sum_H \frac{\#F(H)}{H}. \quad (6)$$

Proof. Each term in $F(H)$ equals $1/H$. The multiplicities remain present in every subsequent infinite sum. $\square$

From now on let $P = \{2, 3, 5\}$ and $S = \mathcal{R}_P$. For $a \geq 0$ define

$$s_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{1}{H(x)}, \quad T_a = \sum_{j \geq 0} s_{a+j}, \quad h_a = \frac{H(2^a)}{2}, \quad X_a = h_a T_a.$$

The half-height h_a clears the finite prefix: for $a \geq 1$ and smooth $x < 2^a$, the exponent of 2 in $H(x)$ is at most $a - 1$, so $H(x) \mid H(2^a)/2$. This is why we omit the final factor 2. Here $h_0 = 1/2$; only for $a \geq 1$ is h_a an integer. Define

$$b_a = \frac{H(2^{a+1})}{H(2^a)}, \quad m_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{H(2^{a+1})}{2H(x)}. \quad (7)$$

Lemma 5.4 (integer coefficients and four possible bases). *For every $a \geq 0$, m_a is a positive integer and $b_a \in \{2, 6, 10, 30\}$. The word “numerator” does not impose the positional-digit restriction $m_a < b_a$; that restriction need not hold.*

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Proof. If $x < 2^{a+1}$, the exponent of 2 in $H(x)$ is at most a , while the other exponents are bounded by those of $H(2^{a+1})$. Hence $2H(x) \mid H(2^{a+1})$. Each summand is an integer, and the shell contains 2^a . Between consecutive powers of 2 there is at most one power of 3 and at most one power of 5: successive powers of either odd prime have ratio greater than two. These factors, when present, and the factor 2 at the right endpoint give the four possible bases. $\square$

In the shell $[2, 4)$ the terms $1/2$ and $1/6$, with $h_2 = b_1 = 6$, give $m_1 = 4$ and the recurrence $X_2 = 6X_1 - 4$. The shell $[16, 32)$ shows why these numerators are not positional digits: $m_4 = 65 > b_4 = 30$. The companion computes the carried digits separately.

Proposition 5.5 (the tail recurrence and a quadratic bound). *The series defining S, T_a converge. For every $a \geq 0$,*

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$$X_{a+1} = b_a X_a - m_a, \quad 0 < X_a \leq \frac{8640}{343}(a+1)^2 < 90(a+1)^2.$$

For every integer $B \geq 1$, either some BX_a is integral and all later states are integral, or $\text{dist}(BX_a, \mathbb{Z}) \geq 1/31$ at arbitrarily large indices.

Proof. For each pair of exponents of 3 and 5, at most one exponent of 2 places the smooth integer in $[2^a, 2^{a+1})$. The two odd-prime exponents lie between 0 and a , giving at most $(a+1)^2$ terms. By (5), $s_a \leq 30(a+1)^2/8^a$. Since $h_a \leq 8^a/2$ and $a+j+1 \leq (a+1)(j+1)$ for $a, j \geq 0$, summing the later shells gives

$$X_a \leq 15(a+1)^2 \sum_{j \geq 0} \frac{(j+1)^2}{8^j} = \frac{8640}{343}(a+1)^2.$$

This proves convergence and the bound. Splitting the first shell gives the recurrence because $m_a = h_{a+1}s_a$ and $h_{a+1} = b_a h_a$. If all sufficiently late distances are strictly below $1/31$, write $BX_a = z_a + e_a$ with $z_a \in \mathbb{Z}$ and $|e_a| < 1/31$. Since Bm_a is integral, $e_{a+1} - b_a e_a$ is an integer of absolute value less than $(1 + b_a)/31 \leq 1$, so $e_{a+1} = b_a e_a$. The lower bound $b_a \geq 2$ forces this bounded error to vanish; integrality then propagates by the recurrence. $\square$

The identity $S/2 = \sum_{a \geq 0} m_a / (b_0 \cdots b_a)$ places the recurrence in the setting of Cantor series. The classical criteria of Erdős–Straus [9, Theorem 2.1] and Hančl–Tijdeman [10, Theorem 3.1] require $m_a / (b_{a-1} b_a) \rightarrow 0$. That hypothesis fails here: the numerators grow quadratically while the bases are bounded. The companion’s Section 7 proves the growth assertion and gives the precise carry conditions. The argument below instead uses the recurrence and the proved bound $90B(a+1)^2$ directly.

Theorem 5.6 (rationality gives positive integer tails). *If $S = A/D$ in lowest terms, where $D = 2^u 3^v 5^w B$ and $\gcd(B, 30) = 1$, then for every $a \geq a_0 = u + 1 + 2v + 3w$,*

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$$d_a = BX_a \in \mathbb{Z}_{>0}, \quad d_{a+1} = b_a d_a - Bm_a, \quad d_a \leq 90B(a+1)^2.$$

Proof. For $a \geq 1$, every $H(x)$ with $x < 2^a$ divides h_a . Subtracting the finite prefix therefore gives

$$X_a = h_a S - \sum_{\substack{x \text{ smooth} \\ x < 2^a}} \frac{h_a}{H(x)}, \quad \sum_{\substack{x \text{ smooth} \\ x < 2^a}} \frac{h_a}{H(x)} \in \mathbb{Z}. \quad (8)$$

The exponent of 2 in h_a is $a - 1$. Moreover $a \geq 2v$ gives $2^a \geq 3^v$, and $a \geq 3w$ gives $2^a \geq 5^w$. Thus $2^u 3^v 5^w \mid h_a$ after the stated onset, and BX_a differs from $h_a A / (2^u 3^v 5^w)$ by an integer and is therefore integral. Positivity, the recurrence and the bound follow from Proposition 5.5. $\square$

Every positive denominator splits in the required way: B is what remains after removing all factors 2, 3 and 5. For example, a hypothetical reduced denominator $2^3 3^{25} 5 \cdot 7$ would give $B = 7$; the next result gives the first integral $7X_a$ at $a = 4$, whereas the sufficient bound above gives $a_0 = 11$.

Corollary 5.7 (the first index at which the denominator clears). *Under the same lowest-terms hypothesis, put $M = 2^u 3^v 5^w$ and let den denote the positive reduced denominator. For $a \geq 1$,*

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$$\text{den}(BX_a) = \frac{M}{\gcd(M, h_a)}.$$

Consequently BX_a is integral exactly when $2^a \geq \max(2^{u+1}, 3^v, 5^w)$. The first such a can be found by integer comparisons, without logarithmic rounding.

Proof. By (8), BX_a differs from $h_a A / M$ by an integer. Since $\gcd(A, M) = 1$, reducing that fraction gives the displayed denominator. The three divisibility conditions for h_a are precisely the three inequalities stated above. $\square$

Where the distinct-height argument stops. The bases b_a of (7) are the numbers $b_{\tau_a} = H(2^{a+1}) / H(2^a)$ of (2), and both sums obey a recurrence of the same shape: $Y_{a+1} = b_a Y_a - \mu_{\tau_a}$ for the distinct-height sum and $X_{a+1} = b_a X_a - m_a$ here. They differ in what is subtracted. The distinct-height sum subtracts one of the five numbers $\mu_\tau \leq 13$, which gives its tails a fixed interval and its maps a finite list. Here m_a counts the smooth numbers of the shell with weights and grows quadratically; the companion’s Section 6 proves $m_a \geq (\lfloor a / (2 \log_2 3) \rfloor + 1)(\lfloor a / (2 \log_2 5) \rfloor + 1)$. Since $X_a > m_a / b_a \geq m_a / 30$, the tails are unbounded, and so are the integers $d_a = BX_a$ that a rational value of S would

produce by Theorem 5.6. The maps $y \mapsto (m_a + y)/b_a$ also change with a , so a repeated value of d_a would not repeat the future of the recurrence. The proof of Theorem 2.6 stops at exactly this step: no finite set of states determines the future of the repeated recurrence, and the rest of this note replaces the return to an earlier state by a comparison of residues.

The two-tail proof of Theorem 2.1 stops at the same place. Over the prime powers, $\mathcal{R}_P = \sum_{k \geq 0} c_k/Q_k$, where c_k counts the smooth integers in $[t_k, t_{k+1})$ (with $t_0 = 1$), and the tails scaled by Q_k are unbounded, as the X_a are. Lemma 2.3 needs every tail in $(0, 1)$, so that an integer equal to a difference of two tails vanishes. With unbounded tails that step fails, and equality of two tails no longer forces agreement on each rearranged run.

The comparison does close on part of the catalogue series. For $p \in P$ let $E_p = \sum_{\alpha \geq 0} L(p^\alpha)^{-1}$ be the sub-sum of $\mathcal{R}_P$ over the powers of p . Then $\mathcal{D}_P = \sum_{q \in P} E_q - (|P|-1)$, and for two primes $\mathcal{R}_{\{p,q\}} = E_p E_q$, which is the factorisation in the proof of Theorem 4.1.

Theorem 5.8 (the single-prime sub-sums). *For every finite set P of primes with $|P| \geq 2$ and every $p \in P$, the number E_p is irrational.*

Ordinary proof in the companion, Section 4.6. It was found within this project on 26 September 2026 and independently refereed by a second reader; it has no Lean proof, so Comparator does not apply. For $|P| = 2$, E_p is an affine function of the Hecke–Mahler value in the proof of Theorem 4.1 and is transcendental; the new case is $|P| \geq 3$, for which we have not searched the literature.

The proof compares the tails after p^α and $p^{\alpha+n}$ as in Section 2, with the invariant replaced by the product of the primes that follow p in a run, and takes the run at a passage of the flow near the origin of the torus; no two-wall lemma is needed. For $\mathcal{R}_P$ itself the comparison fails at its first step: between the two scales the tail of $\mathcal{R}_P$ gains the early terms of every ray $p^\alpha w$, with w a $(P \setminus \{p\})$ -smooth integer, that enters there, and E_p is the case $w = 1$, where nothing enters.

6 A window test and the remaining arithmetic

Two steps of the recurrence already show the finite data we need:

$$X_{\ell+2} = b_{\ell+1} b_\ell X_\ell - (b_{\ell+1} m_\ell + m_{\ell+1}).$$

If BX_ℓ is an integer, this identity determines the residue of $BX_{\ell+2}$ modulo the product of the two bases. We will compare the least positive representative of that residue with the upper bound for the endpoint tail.

For a window starting at $\ell \geq 0$ and having $h \geq 0$ steps, define its product of bases and accumulated numerator from (7) by

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \quad W_{\ell,h+1} = b_{\ell+h} W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h} F_{\ell,h} + m_{\ell+h}.$$

Induction on h gives

$$X_{\ell+h} = W_{\ell,h} X_\ell - F_{\ell,h}. \tag{9}$$

For integers $W \geq 1$ and t , define $\text{lpr}_W(t) = 1 + ((t-1) \bmod W)$, using a remainder in $\{0, \dots, W-1\}$. Thus a zero residue is represented by W , not 0. Throughout this section use the upper bound

$$K(B, a) = 90B(a+1)^2. \tag{10}$$

Lemma 6.1 (least positive residues). *If d is a positive integer with $d \leq K$ and $d \equiv -BF \pmod{W}$, where $W \geq 1$, then $\text{lpr}_W(-BF) \leq K$.*

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Proof. Every positive representative of the residue is at least its least positive representative, so $\text{lpr}_W(-BF) \leq d \leq K$. $\square$

For example, the window starting at $\ell = 1$ with length 6 has $W_{1,6} = 648000$ and $F_{1,6} = 524431$. At $B = 1$ its least positive residue is 123569, exceeding $K(1,7) = 5760$. This rules out an integral X_1 , but not a rational denominator that would clear only at a later index. The theorem therefore asks for windows beyond every prescribed starting index, as well as for every eligible denominator.

Theorem 6.2 (a residue criterion for irrationality). *The number S is irrational if and only if*

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$$\begin{aligned} &\text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a_0 \geq 1, \\ &\text{there are } \ell \geq a_0, h \geq 1 \text{ such that } \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > K(B, \ell + h). \end{aligned} \quad (11)$$

Proof. Assume (11) and suppose S is rational. Theorem 5.6 supplies a positive integral carry $d_a = BX_a$ after its onset. Choose a window in (11) beyond that onset. Equation (9) gives $d_{\ell+h} \equiv -BF_{\ell,h} \pmod{W_{\ell,h}}$, contradicting Lemma 6.1 and the upper bound.

Conversely, let S be irrational and fix $B, \ell \geq 1$. Equation (8) makes BX_ℓ nonintegral. Put $\delta = \lceil BX_\ell \rceil - BX_\ell \in (0, 1)$. Since $W_{\ell,h} \geq 2^h$ and $X_{\ell+h} = O((\ell + h + 1)^2)$, for all sufficiently large h the number $\delta + BX_{\ell+h}/W_{\ell,h}$ lies in $(0, 1)$. The window identity shows that the integer $\lceil BX_\ell \rceil W_{\ell,h} - BF_{\ell,h}$ lies strictly between 0 and $W_{\ell,h}$ and is congruent to $-BF_{\ell,h}$. Hence

$$\text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \delta W_{\ell,h} + BX_{\ell+h}.$$

Its exponentially growing first term eventually exceeds $K(B, \ell + h)$. This proves (11), in fact from each fixed start. $\square$

The proved bound $90B(a+1)^2$ controls every positive integral BX_a that rationality would force. Polynomial growth alone does not suffice: the zero bound would make the residue inequality automatic. The companion, Section 8, treats smaller and more general bounds. No finite search establishes the unbounded quantifiers in (11).

The remaining task has the following three equivalent formulations.

Problem 6.3 (the repeated three-prime target). Prove that $S = \mathcal{R}_{\{2,3,5\}}$ is irrational.

Problem 6.4 (no reduced tail is an integer). Prove that for every $a \geq 1$ and every integer $B \geq 1$ coprime to 30,

$$BX_a \notin \mathbb{Z}. \quad (12)$$

Problem 6.5 (residue inequalities after every starting index). Prove (11).

Indeed, one integral scaled tail makes S rational by (8), whereas denominator clearing gives such tails from a rational S . Theorem 6.2 supplies the third equivalence. These arithmetic assertions remain unproved. The companion's finite certificates (Section 9) and its comparisons with other approaches (Section 10) are supplementary; the argument here does not rely on them.

Statements and declarations

Proof sources. Theorem 4.1 uses the transcendence theorem of Bugeaud and Laurent, which its Lean proof assumes. Fan’s post of 26 June 2026 on the erdosproblems.com forum has priority for the two-prime deduction [7]. Theorem 2.1 and Lemmas 2.2–2.5 have the ordinary proofs given in Section 2 and no Lean proofs; a formal proof would need Kronecker’s theorem on the closure of a line in a torus and the Euler characteristic count of Lemma 2.5. The proof of Theorem 2.1 was found by an AI agent on 26 September 2026 and refereed by a second AI agent working independently, who re-derived every step and closed five small gaps; it has had no human review. Theorem 5.8 has the ordinary proof in the companion, refereed in the same way, and no Lean proof. Theorem 2.6 is checked in Lean, as the remark below it states; Comparator has not yet compared it. Corollary 2.7 has an ordinary proof together with an exact finite computation, and no Lean proof. Every other theorem, proposition, lemma and corollary has a Lean proof of its whole statement, marked in the margin; Appendix A describes the formal sources. Neither these proofs nor the finite calculations supply the required windows for every eligible denominator and arbitrarily late starts.

Artefact and data availability. The companion contains the detailed bounds and supplementary arguments; its source inventory explains the normalisation conventions of the Lean sources. The [distinct-height programs](#) recompute the constants of Theorem 2.6 and the certificates behind Corollary 2.7 and the four-prime data in exact rational arithmetic, and store their outputs; the same directory holds exact illustrations of the proof of Theorem 2.1, which the proof does not use. AI agents carried out most of the research and drafting. The work has not had independent human mathematical review.

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A Guide to the formal sources

There are four separate arguments. The proof of Theorem 2.1 uses integral tails, rearranged blocks, a closed subtorus and an Euler characteristic count, and has no Lean proof. The proof of Theorem 2.6 uses dyadic blocks, the disjoint images of five affine maps and the irrationality of $\log 2 / \log 3$; its Lean proof follows the same route. The rank proof uses the height formula, row and column rescaling, and a threshold determinant. The arithmetic argument uses the shell coefficients, the tail recurrence, denominator clearing and least positive residues. The Lean proof of Theorem 6.2 establishes the equivalence of the residue condition with irrationality; it does not establish the residue condition itself.

Library declarations. The margin marks link each result to its Lean proof. The main steps rest on the following library declarations, each opened at a fixed revision. Distinct heights: the height is the running LCM; the sum counts each distinct value once; a power of 5 in every three consecutive blocks; the blocks containing a power of 3 are not eventually periodic; the criterion for the five block letters; the irrationality of $\mathcal{D}_{\{2,3,5\}}$. Running LCM: the exact height formula; constancy on each logarithmic cell; the jumps by p , q and r ; exactly 3 count + 1 jump values; the quadratic bound on exponent shells. Dyadic blocks: at most one internal pure-power jump per channel; the radix is 2, 6, 10 or 30; the radix lies in $[2, 30]$. Kernel rank: the kernel is not of rank one; the smallest 2×2 minor equals $-1/15$; the tail-state recurrence. Residues and carries: no bounded positive state represents an escaping residue (integer form); the local-window escape condition and its exclusion of a positive reduced carry; a bounded-radix orbit is integral or cofinally far from the integers; no uniformly bounded nonzero carry lift; block-null words are channel coboundaries; the carry as a residue digit plus a coboundary; a finite-dimensional factorisation gives a finite realised span.

Scope of the supplementary literature. The companion develops the comparisons omitted from the main argument. Its Section 6 relates the bounded-ratio argument to Erdős–Taylor [11, Theorem 1, p. 600] and Fan [12, Lemma 3.1, p. 7]; Section 7 compares the exact carry restrictions and polynomial-radix assumptions of Hančl–Tijdeman [13, Theorems 2.2, 3.1 and 4.2]. The Isabelle development of Koutsoukou–Argyragi and Li [19] verifies classical criteria, not the present series.

Section 10 of the companion separates the finite-difference hypotheses of Luca–Ouaknine–Worrell [14, Definition 5, Theorems 6 and 8, Claim 10] from quadratic growth alone, and checks the obstacles to using the echoing [15], digit-complexity [20, Theorem 1] and multivariate Mahler [21] results. These comparisons do not establish the missing hypotheses for S .

For broader background, the companion distinguishes work on fixed-prime semi-groups [16, 17] from other unit-fraction irrationality problems [18]. No estimate from those papers is used in this note’s counting argument.

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